MILITARY SPECIFICATION

CONTROL ASSEMBLY , PUSH-PULL, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1, SCOPE

1.1 Scope. - This specification covers push-pull control assemblies for general applications used to regulate and actuate internal combustion engine and vehicular accessories (see 6.1).

1.2 Classification - Control assemblies shall be of the following types as specified (see 6.2).

Type I - Friction Lock
Type II - Positive or mechanical lock
Type III - Vernier adjustment

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-489 - Enamel, Alkyd, Gloss, Synthetic (for Exterior and Interior Surfaces)
PP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PP-B-636 - Box, Fiberboard
MIL-P-116 - Preservation, Methods of
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage
MIL-STD-l30 Identification Marking of U. S. Military Property
MS 500043 Control Assembly, Push-Pull, Throttle, Engine, Ratchet Lock Type

FSC 2990

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MS500044	Control Assembly, Push-Pull, Throttle, Engine, Vernier, Adjustment Type
MS500045	Control Assembly, Push-Pull, Choke Engine
MS500073	Control Assembly, Push-pull, Friction Lock Type, General Purpose
MS500074	Control Assembly, Push-Pull, Ratchet Lock Type, General Purpose

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other Publications - The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw Thread Standards for Federal Service

Department of Transportation

Federal Motor Vehicle Safety Standard 201, Amended, Occupant Protection in Interior Impact

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Preproduction Sample - When specified (see 4.2 and 6.2) a preproduction sample of the control assemblies to be furnished under this specification shall be produced prior to the manufacture of these items in production quantities. Under the surveillance of Government representatives, designated by the contracting officer, the manufacturer shall subject this assembly to the preproduction inspection specified herein to determine conformance to the quality assurance provisions of this specification. The preproduction sample shall be fully representative of the assemblies to be supplied from production facilities and tooling.

3.2 Materials - Materials shall be as specified herein and on applicable drawings, standards and specifications. When the material used is not specifically covered in any of the reference documents, it shall withstand the temperature range encountered within the assembly without any change in physical or chemical properties affecting performance of the control.

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3.3 Design and Construction - Control assemblies shall be in accordante with applicable configuration shown in Figure I and Table 1, or drawings, standards or other procurement documentation as specified (see Fig. I, 6.1.1 and 6.2). Physical configuration, other than set forth in this specification or as specified on applicable procurement documentation, shall be at the option of the manufacturer. Design requirements specified on standards, or drawings shall take precedence over requirements specified herein. In the event of a conflict between a Purchase Description and the data listed in Table I for a specific Federal Stock Number the Purchase Description shall take precedence.

3.3.1 Standard Parts - Military standard parts shall be utilized wherever possible.

3.3.2 Threaded Parts - All screw threads shall conform to Handbook H28 . The form, number per inch and class of threads shall be as specified on applicable procurement documentation (see 3.3).

3.3.3 Finish - All exposed metallic surfaces shall be of corrosion resistant materials or shall be plated for corrosion resistance.

3.3.3.1 Painting - When specified on applicable procurement documentation (ace 3.3) the collar nut (cover cover) and hand grip shall be painted olive drab.

3.3.4 Effective Length - The minimum effective length of the control assembly shall be as shown on specified procurement documentation (see 3.3).

3.3.5 Travel.

3.3.5.1 Type I - Minimum travel shall be as shown on specified procurement documentation (see 3.3).

3.3.5.2 Type II - The minimum locking travel shall be as shown on specified procurement docementation (see 3.3).

3.3.5.3 Type III - The minimum vernier adjustment travel length shall be as shown on specified procurement documentation (see 3.3).

3.3.6 Exposed Core Length - The minimum exposed core length shall be as shown on specified procurement documentation (see 3.3).

3.3.7 Hand Grip - A hand grip shall be provided for actuating the control. The hand grip shall be either T-handle or knob style as shown in procurement documentation (see 3.3).

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TABLE I - PUSH PULL CONTROL CONSTRUCTION - SEE FIGURE I FOR DIMENSIONAL FINDS AND GENERAL CONFIGURATION

DASH NO.	FEDERAL STK. NO.	HANDLE IMPRINT	HANDLE STYLE	A	B	C	D	WIRE DIA.	E TRAVEL (MIN)	CONTROL TYPE
1	2990-990-1304	CHOKE	1	.375-24	.500	163.000	166.000	.054	2.250	FRICTION
2	2990-838-2590	CHOKE	1	.375-24	.500	96.000	98.000	.054	2.250	FRICTION
3	2990-590-2666	CHOKE	2	.375-24	1.500	60.000	65.000	.054	3.000	FRICTION
4	2590-822-7978	CHOKE	1	.375-24	1.827	127.000	129.500	.072	3.000	FRICTION
5	2590-996-7284	CHOKE	1	.437-20	1.750	92.000	94.500	.054	2.250	FRICTION
6	2590-318-0491	CHOKE	1	.375-24	.468	296.500	299.000	.054	2.500	FRICTION
7	2590-289-8705	THROTTLE	2	.375-24	1.437	86.312	89.312	.054	3.000	FRICTION
8	2990-822-3283	THROTTLE	1	.375-24	.375	34.000	38.000	.054	2.250	FRICTION
9	2910-768-7370	T	1	.375-24	.500	53.125	55.750	.047	2.250	FRICTION
10	2990-774-5571	NOT MARKED	1	.375-24	.500	140.500	144.000	.054	2.750	FRICTION
11	2990-273-7139	NOT MARKED	2	.375-24	.531	88.000	90.000	.055	2.250	FRICTION
12	2990-930-5693	NOT MARKED	2	.375-24	.500	42.000	45.500	.054	2.750	FRICTION
13	2590-540-6524	NOT MARKED	2	.500-20	1.187	108.000	113.000	.090	4.500	FRICTION
14	2590-798-1056	NOT MARKED	1	.375-24	1.500	55.250	58.250	.072	2.750	FRICTION
15	2590-711-5824	NOT MARKED	1	.375-24	1.438	28.000	30.750	.062	2.750	FRICTION
16	2590-273-8832	NOT MARKED	2	.437-20	2.000	28.000	32.000	.072	2.500	FRICTION
17	2990-204-3179	NOT MARKED	2	.500-20	.750	216.000	221.000	.062	3.000	FRICTION
18	2590-287-7466	DETENT	2	.375-24	1.500	173.375	175.187	.054	2.250	FRICTION
19	2990-690-9975	DETENT	2	.375-24	1.500	72.000	73.812	.054	2.250	FRICTION
20	2590-980-3797	ENG. STOP	2	.375-24	1.375	99.000	102.000	.054	2.000	LOCK
21	2990-131-6151	EMERGENCY ENGINE STOP	2	.375-24	1.438	59.750	66.125	.054	3.000	LOCK
22	2590-410-2263	V	1	.375-24	.625	32.000	33.750	.054	2.250	FRICTION
23	2990-915-6354	QUICK START	1	.375-16	.750	34.000	37.000	.054	2.750	FRICTION
24	2990-702-3048	THROTTLE	2	.375-24	1.500	120.000	122.000	.054	2.250	LOCK
25	2990-580-7379	THROTTLE	2	.375-24	1.500	79.000	84.000	.054	2.250	LOCK
26	2590-905-4428	THROTTLE	2	.437-24	2.000	146.000	150.000	.072	2.500	LOCK
27	2990-407-7791	THROTTLE	2	.437-20	2.000	32.000	36.000	.054	3.000	LOCK
28	2590-039-8916	AUX. CHOKE	2	.500-20	1.500	34.000	37.500	.072	3.000	LOCK
29	2990-294-6472	NOT MARKED	2	.375-24	1.500	88.000	92.000	.054	3.000	LOCK
30	2590-580-9055	NOT MARKED	1	.375-24	.375	22.000	24.250	.054	2.250	LOCK

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TABLE I - PUSH PULL CONTROL CONSTRUCTION - SEE FIGURE I FOR DIMENSIONAL FINDS AND GENERAL CONFIGURATION

DASH NO.	FEDERAL STK. NO.	HANDLE IMPRINT	HANDLE STYLE	A	B	C	D	WIRE DIA.	E TRAVEL (MIN)	CONTROL TYPE
31	2590-019-4568	NOT MARKED	2	.437-20	2.000	132.000	135.750	.054	3.000	LOCK
32	2990-896-2166	NOT MARKED	2	.437-20	1.750	72.000	88.000	.054	3.000	LOCK
33	2990-678-9664	NOT MARKED	2	.437-20	1.562	61.000	67.000	.072	.750	LOCK
34	2910-698-0384	NOT MARKED	2	.437-24	2.000	33.750	38.500	.072	2.500	LOCK
35	2590-772-1799	NOT MARKED	2	.500-20	.812	194.250	198.250	.072	3.000	LOCK
36	2990-318-0945	NOT MARKED	2	.375-24	1.500	114.000	119.500	.072	3.000	LOCK
37	2590-828-5342	TURN TO LOCK	1	.562-18	.750	83.000	89.000	.054	3.500	LOCK
38	2990-392-4417	NOT MARKED	3	.750-16	2.500	144.000	147.000	.080	3.250	VERNIER
39	2990-992-4429	NOT MARKED	3	.750-16	2.375	63.000	69.250	.080	4.000	VERNIER
40	2990-640-3853	NOT MARKED	3	.750-16	2.562	113.500	118.000	.062	3.000	VERNIER
41	2990-931-2231	NOT MARKED	3	.750-16	2.562	72.000	76.000	.062	3.000	VERNIER

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3.3.7.1 T-Handle - T-handle grip shall be of a nominal 2 inch overall length, smooth and free from burrs or slivers. Handle shall be made from molded plastic, zinc alloy casting, or chrome plated steel.

3.3.7.2 Knob Handle - Knob grip shall be of 7/8 inch minimum diameter smooth and free from burrs or slivers. Handle shall be made from molded plastic, zinc alloy casting, or chrome plated steel.

3.3.7.3 Handle Marking -When marking is specified in procurement documentation (see 3.3), it shall be permanent and indented into the handle, with not less than 1/8 inch size figures and be marked with enamel conforming to Federal Specification TT-E-489.

3.3.8 Core - Core wire shall be of solid one piece or stranded construction and shall be made of corrosion-resistant steel or of steel furnished with a corrosion-resistant electrochemical finish. Core diameter shall be as shown in procurement documentation (see 3.3).

3.3.9 Conduit - Unless otherwise specified (see 6.2), conduit shall be 0.187 inch minimum outside diameter, composed of 0.054 inch minimum diameter steel wire. galvanized finish, formed in a closed helix. When specified (see 6.2), conduit shall be encased in a plastic or rubber outer sleeve.

3.3.9.1 Inner Lining - When specified (see 6.2), conduit shall be provided with a plastic inner lining.

3.3.10 Body or Guide Tube - Guide tube shall be of steel with corrosion-resistant, electrochemical finish, and shall usually incorporate the threaded mounting adapter and mounting shoulder. When specified (see 6.2), unlined guide tubes shall be assembled pre-lubricated with grease conforming to specification MIL-G-23827 or equivalent.

3.3.10.1 Mounting Adapter - The body tube shall provide a threaded mounting adapter. Thread size and minimum length shall be as specified in applicable procurement documentation (see 3.3). Mountnig hardware, consisting of nut and lockwasher shall be provided with each control assembly.

3.3.10.2 Mounting Shoulder (Face Plate or Cover Plate)- Unless otherwise specified (see 6.2), a fixed mounting shoulder or cover plate shall be incorporated on the body tube of adequate diameter to facilitate mounting of the assembly to opening through control panel or dash board, Outside diameter of mounting shoulder shall be not less than 1/4 inch diameter greater than the diameter of the threaded adapter for Type II and Type III controls and not less than 1/8 inch diameter greater for Type I controls.

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3.3.11 Plunger Rod - Handle shall be affixed to a plunger rod which shall actuate back and forth inside of the body or guide tube snugly, but freely without binding. When installed, the handle shall not rotate or swivel on the rod. Rod shall be of corrosion-resistant steel, or of steel furnished with a corrosion-resistant electrochemical finish. On Type I assemblies, the rod shall be constructed so as not to swivel or turn inside the guide tube.

3.4 Performance

3.4.1 Locking Devices

3.4.1.1 Positive (Type II) - The Type II assemblies shall be capable of locking through the complete travel range of 3 inches with locking position adjustments to be possible at not more than 1/8 inch internals. Locking and unlocking shall be accomplished with a minimum of 1/4 turn and a maximum of 3/4 turn of the handle to right or left, Control shall operate freely throughout its entire travel length smoothly without binding.

3.4.1.2 Vernier Adjustment (Type III) - Type III assemblies shall be capable of fine adjusting control settings at a infinite number of positions throughout the replete travel range. in addition, a push button release mechanism shall be incorporated for releasing the vernier adjustment to provide for coarse adjustment of the control position throughout the entire travel range prior to vernier adjustment. When the coarse adjustment button is released, the vernier adjustment control shall automatically become operable. Control shall operate smoothly throughout its entire travel length without binding.

3.4.1.3 Friction Lock (Type I) - The Type I assembly shall be of such an inherent design that a force of 3 to 8 pounds is required to actuate the assembly to adjust position of the control. The assembly shall operate smoothly through-out its entire travel length without binding.

3.4.2 Lock Holding Ability (Type II and III) - Type II and III controls shall be able to withstand a cycling load specified in paragraph 4.5.2 without change of pre-set control adjustment position, or without damage to control affecting operation

3.4.3 Maximum Pull Load

3.4.3.1 Type I - The control assembly shall be capable of withstanding a steady, gradually applied, pull load of 120 pounds applied to the control handle without separation of the handle from the assembly or other damage to the control affecting performance of the assembly.

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3.3.11 Type II and III - Unless otherwise specified (see 6.2), the control assembly shall be capable of withstanding a gradually applied steady pull load of 751 pounds applied to the-handle without separation of the handle from the assembly or other damage affecting performance.

3.4.4 Bend Radius - The Types I and II control assembly shall function smoothly with a conduit bend radius of 3 inches with a maximum push-pull actuating force of 20 pounds, while the Type III control assembly shall be capable of vernier adjustment with a maximum force of 10 inch - lbs. The control assembly shall show no permanent set as a result of this radius bend.

3.4.5 Temperature Range Extremes - The control assembly shall be capable of operating satisfactorily in a temperature range of from minus 65°F to plus 13.5 F, with maximum push-pull actuating force of 30 pounds.

3.5 Workmanship - The finished product shall be of first class commercial workmanship, free from burrs, slivers, surface scratches, sharp corners or edges, paint runs or smears.

3.6 Identification Marking - Marking shall consist of MS part number if applicable, manufacturers name and identification number, in accordante with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Inspection - When required (see 3.1 and 6.2), a preproduction sample shall be furnished. Preproduction inspection and testing shall be performed by the contractor under Government surveillance and shall consist of examination for defects specified in 4.3.2.2 and performance of tests specified in Table II, in the order listed.

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TABLE II - ORDER OF PREPRODUCTION TESTING

TEST	REQUIREMENT	PARAGRAPH NO,.
Locking Device Performance	3.4.1.1, 3.4.1.2, 3.4.1.3	4.5.1
Maximum Load	3.4*3	4.5.3
Lock Holding Ability	3.4.2	4.5.2
Extreme Temperature	3.4.5	4.5.4
Bend Radius	3.4.4	4.5.5
Locking-Device Performance	3.4.1.1, 3.4.1.2, 3.4.1.3	4.5.1

4.2.1 Failure - Failure of a preproduction sample to pass any of the examinations or teats specified herein, may be cause for the Governmen t's refusal to witness additional examinations or tests until the faults revealed have been corrected,

4.3 **Quality Conformance Inspection**

4.3.1 Sampling

4.3.1.1 Lot Fomation - A lot shall consist of all controls of one type, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for Examination - Samples for quality Conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for Testing - A representative sample shall be selected in accordance with inspection level S-1 of MIL-STD-105 from each lot that has passed the examination specified in 4.3.2.2.

4.3.2 Examination

4.3.2.1 Acceptable Quality Level - Each control assembly selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classificatin	AQL
Major	1.0
Minor	2.5

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4.3.2.2 Classification of Defects - For examination purposes, defects shall be classified as follows:

Major Defects (1.0 AQL)	Examination Method
Effective length less than specified (see 3.3.4)	SIE*
Travel less than specified (see 3.3.5)	SIE*
Exposed core length less than specified (see 3.3.6)	SIE*
Handle marking affecting operation (see 3.3.7.3)	Visual
Incorrect dimensions affecting interchangeability (see 3.3)	SIE*
Handle dimensions less than specified (see 3.3.7)	SIE*
Core wire material or size not as specified (see 3.3.8)	SIE*
Guide tube mounting facilities not provided (see 3.3.10)	visual
Conduit size or material not as specified (see 3.3.9)	SIE*
Cover plate size not adequate for mounting (see 3.3.10.2)	SIE*
Plunger rod material or design not as specified (see 3.3.11)	Visual, SIE*
Minor Defects (2.5 AQL)	
Incorrect dimensions. not affecting interchangeability (see 3.3 thru 3.3.11)	SIE*
Improper identification of product (see 3.6)	visual
Poor workmanship (see 3.5)	Visual
Handle marking not affecting operation (see 3.3.7.3)	Visual

*SIE - Standard inspection equipment

4.3.3 Quality Conformance Tests - Samples selected in accordance with 4.3.1.3 shall be subjected to the test specified in 4.5.5, and 4.5.3 and 4.5.1 using an AQL of 6.5 percent defective.

4,3,4 Rejection and Retest - Control assemblies that have been rejected may be reworked or have parts replaced to correct defects and be resubmitted for tests. Before resubmitting, detailed information or data concerning the rejection and the action taken to correct the defects shall be furnished to the Government inspector.

4.4 Control Tests

4.4.1 Sampling - control test samples shall be selected at the rate of one of each 500 units produced, except that not less than one or more than three units shall be selected in any 30-day period. Control assemblies shall be examined for the defects specified in 4.3.2.2, and subsequently tested as specified in 4.5.5, 4.5.3 and 4.5.1.

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4.4.2 Failure - Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the examination or test have been corrected.

4.5 Test Procedure -Unless otherwise specified herein, tests shall be conducted under the following conditions:

The control assembly shall be mounted in a position to simulate its actual application and a load of one pound plus or minus 3 ounces affixed to the exposed core end.

4.5.1 Locking Device Operation

4.5.1.1 Type II - The Type II control assembly shall be actuated through its complete trowel length and locked at not less than eight different settings to determine conformance to paragraph 3.4.1.1.

4.5.1.2 Type III - To determine conformance to paragraph 3.4.1.2 the Type III control assembly shall be actuated by use of the vernier adjustment throughout its entiere travel length. In addition the push button coarse adjustment mechanism shall be actuated and the control moved through itws entire length of travel. Upon release of the button the vernier adjustment shall again be operable.

4.5.1.3 Type I - To determine conformance to paragraph 3.4.1.3, the control assembly shall be actuated, with no load attached, through its entire length of travel. The force required to actuate the assembly shall be not les then 3 pounds, or not greater than 8 pounds.

4.5.2 Lock Position Holding Ability Type II and III - The control assembly shall be mounted as specified in paragraph 4.5, except the amount of load applied to the exposed core shall be eight ounces. The control shall be locked in an approximate mid point position of travel, and the length of exposed core measured to the nearest 1/32 inch with a pull load of eight ounces on the core. The eight ounce load shall then be applied momentarily in alternating directions along the horizontal axis of the core away from and towards the control head for a complete test cycle of 50 times in each direction. The exposed core length shall then be measured again with the eight ounce pull load and shall not vary by more than 1/32 inch from the orignal length. In addition, the control assembly shall be subjected to locking device operation test cited in 4.5.1 to determine that no damage has incurred affecting operational characters tics.

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4.5.3 Maximum Load - The control assembly shall be mounted in a simulated operating position with the exposed core firmly held in place and a steady load applied to the control handle, increasing until a value of 115 to 120 pounds is reached for the Type I controls and 70 to 75 pounds for the Type II and III controls. Upon release of this load, the control assembly shall be examined for evidence of damage, such as separation of handle from the assembly, and tested for performance as specified in 4.5.1.

4.5.4 Temperature Operation Range

4.5.4.1 Low Temperature - The control assembly shall be subjected to a temperature of minus 60°F to 65°F until stabilized, and while at that temperature shall be actuated through its entire range of travel. The vernier adjustment for Type III controls shall be operable, and the force required to actuate the control for Type I and II assemblies, and the coarse adjustment for Type III assembly, shall be recorded to determine conformance to paragraph 3.4.5.

4.5.4,2 High Temperature - The test specified in paragraph 4.5.4.1 shall be repeated to determine conformance to paragraph 3.4.4 except the temperature shall be plus 110°F to 115°F.

4.5.4.3 Performance After Temperature Extremes - At the conclusion of the temperature extreme tests specified, the cable assembly shall be subjected to visual examination for permanent damage to its components and subjected to performance tests cited in paragraph 4.5.1.

4.5.5 Bend Radius - The control assemblies shall be mounted as for normal operation except the conduit shall be contoured to a 90° bend radius of 3 inches. With M load on the exposed core, the control shall be actuated throughout its entire travel range with force required for actuation recorded to determine conformance to paragraph 3.4.4.

4.6 Inspection of Preparation for Delivery

4.6.1 Materials and Recesses - The Government inspector shall at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of Section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Presentation and Packaging - Preservation and packaging shall be A or C as specified (see 6.2).

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5.1.1 Level A

5.1.1.1 Cleaning and Drying - Items shall be cleaned and dried in accordance with the requirements of MIL-P-116.

5.1.1.2 Packaging - Each item, cleaned and dried as specified, shall be individually protected in accordance with Method IC-2 of MIL-P-116, using a snug fitting box or carton of selected material or construction subject to weight and size and other limitations of the box specification.

5.1.1.3 Intermediate Packaging - Except as otherwise specified herein, unit packages shall be intermediate packaged. in snug . fitting boxes conforming to PPP-B-636, class weather-resistant. Boxes shall be closed with tape. Boxes shall not exceed 40 pounds gross weight or 1.5 cubic feet. Intermediate packaging is not required when the size or weight of the unit package would result in less than two unit packages per intermediate container, or when the total shipment to one destination would result in only one intermediate container per shipping container.

5.1.2 Level C - The items shall be cleaned and dried and packaged in a manner that will afford protection against corrosion, deterioration mechanical and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his own practice providing it fulfills these requirements.

5.2 Packing - Packing shall be Level A, B, or C as specified.

5.2.1 Level A - Unit and intermediate packages shall be overpacked in wood boxes conforming to PP-B-621, class 2, style optional. When the gross weight exceeds 250 pounds, the shipping container shall have skids applied in accordance with the applicable box specification.

5.2.2 Level B - Unit and intermediate packages shall be overpacked in class weather-resistant fiberboard boxes conforming to PPP-B-636, or in wood boxes conforming to PPP-B-621, classs domestic, style optional. When the gross weight exceeds 250 pounds, the shipping container shall have skids applied in accordance with applicable box specification.

5.2.3 Level C - Unit and intermediate packages shall be overpacked in such a manner which affords protection at the lowest rate against damage during direct shipment from the supply source to the first receiving activity. This level shall conform to applicable carrier rules and regulation and may be the suppliers commercial practice providing it meets these requirements.

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5.3 Marking - In addition to any special markings required by the contract or order, unit, intermediate packages, and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended Use - This specification is intended to include common push-pull controls of the single core wire type used basically for control and adjustment of internal combustion engine accessories such as throttles, chokes and governors. They consist primarily of a control handle affixed to plunger rod to which a core wire is permanently attached. This core wire is actuated forwards and backwards inside of a flexible conduit. The handle end of the flexible conduit normally is attached to a body or guide tube in which the plunger rod moves and also incorporates a threaded mounting adapter, a cover plate or mounting shoulder, and mounting hardware to pemit mounting of the assembly through a hole in the dashboard or control panel. The far end of the core wire extends beyond the end of the conduit to permit attachment to the accessory which is to be regulated.

6.1.1 Use of Existing Configurations - In order to keep the number of Federal Stock Numbers in the supply system to a minimum, it is recommnended that the type of control required be determined, and the existing configurations shown on referenced military standards and in Table 1 be reviewed. Where requirements exist for a type of control which is included in the standards or table, except for an effective length difference, the closest longer length indicated should be ordered. The control may then be simply cut to the desired length required for a specific installation.

6.2 Ordering Data - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type of control (see 1.2).

c. Preproduction testing if required (see 3.1).

d. Design and construction document reference (see 3.3).

e, Conduit if different, and outer covering, if required, (see 3.3.9).

f. Inner lining, when required (see 3.3.9.1).

g. Lubricated control, when required (see 3.3.10).

h. Mounting shoulder if different (see 3.3.10.2).

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i. Maximum pull load if different (see 3.4.3.2).

j. Responsibility for inspection, if different (see 4.1).

k. Preservation, packing, packaging and marking levels (awe 5.1).

6.3 Definitions

6.3.1 Effective Length - The distance from the back of the mounting shoulder or cover plate, to the end of the flexible conduit. Does not include exposed core length.

6.3.2 Exposed Core - The length of core wire that extends beyond the end of the conduit when the assembly is In the closed or shut position.

6.4 Safety Standard Compliance - Use of the push-pull control assembly in a motor vehicle may not comply with Department of Transportation, Federal Motor Vehicle Safety Standard 201, amended, Occupant Protecction in Interior Impact.

CUSTODIANS:

Army - AT
Navy - YD
Air Force - 82

Army - AT

Project Number: 2990-0065b

Review Activity:

Navy - YD
DSA - CS
Army - GL

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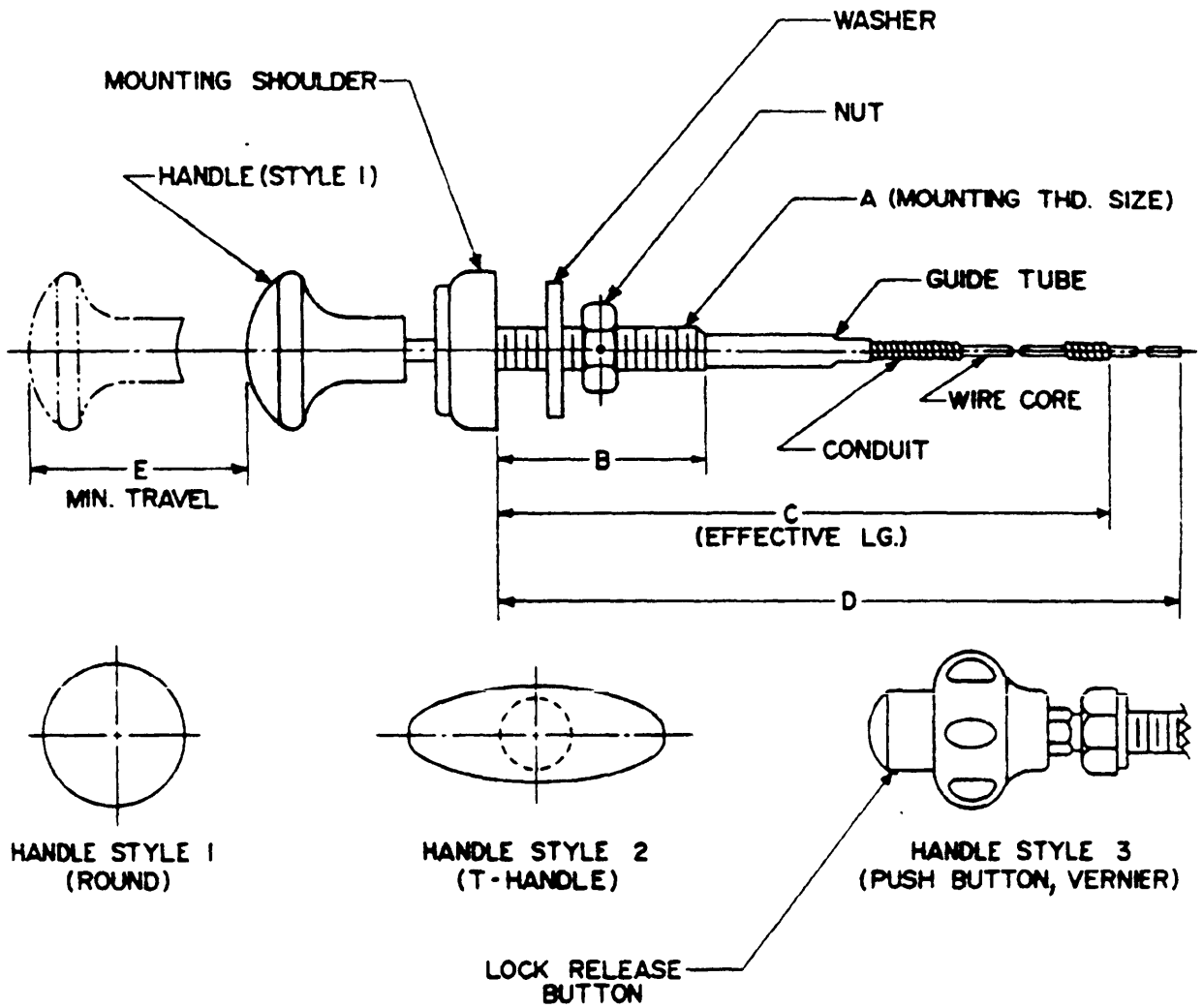


FIGURE 1 - PUSH PULL CONTROL

TYPICAL CONFIGURATION - SEE TABLE I
FOR DIMENSIONAL LIMITATIONS

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FOLD

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

Commander
US Army Tank-Automotive Command
ATTN: AMSTA-RSES
Warren, MI 48090

FOLD

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-C-62191

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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